

Review: Simple Linear Regression

Presentation Outline

- 1 Introduction
- 2 Example
- 3 How it works
- 4 Interpretation
- 5 Predictions and Residuals
- 6 Important points

What Is Simple Linear Regression?

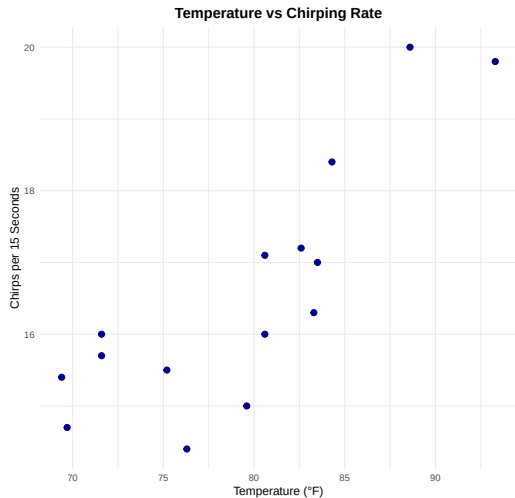
- Models the relationship between two **quantitative** variables
- Predict a **response variable** Y from an **explanatory variable** X
- Model form:

$$Y = b_0 + b_1 X$$

- b_0 : Intercept
- b_1 : Slope

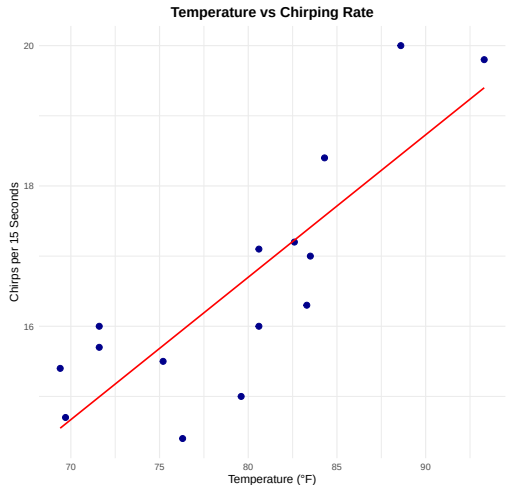
Cricket Chirps and Temperature

- Data on number of chirps (per 15 sec) and temperature (°F)
- Each point: one observation of chirping rate and ambient temperature
- Positive trend: more chirps $\Rightarrow$ higher temperature



What Does the Regression Line Do?

- Best-fitting line through the data points
- Minimizes the **sum of squared vertical distances**
- Called **least squares** regression



Interpreting Coefficients

Example Equation

$$\text{Chirps} = -0.29 + 0.23 \times \text{Temperature}$$

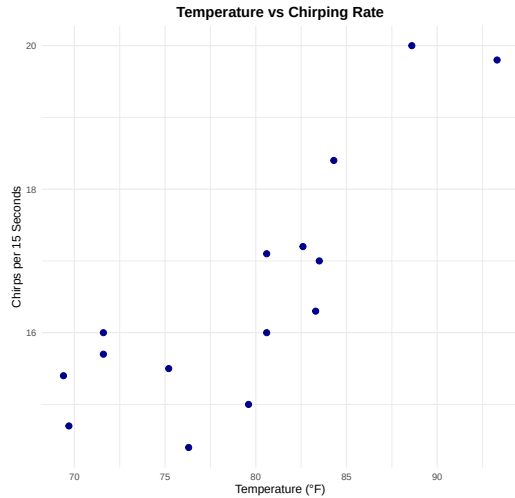
- **Slope (0.23):** For each 1°F increase in temperature, chirping rate increases by 0.23 chirps per 15 sec
- **Intercept (-0.29):** Predicted chirping rate when temperature = 0°F (not meaningful in context)

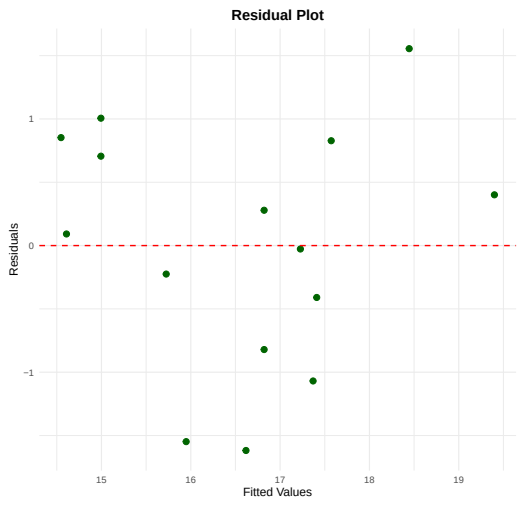
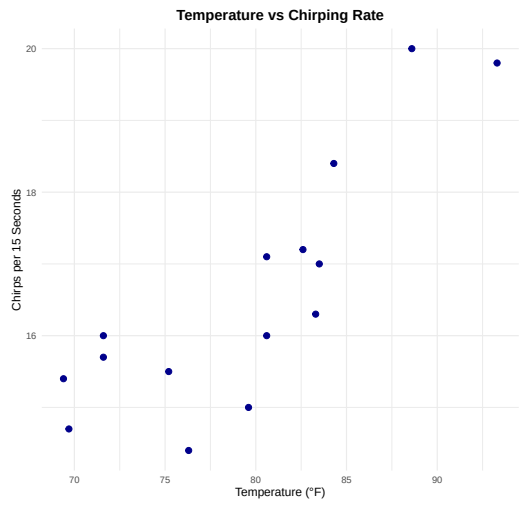
Predictions and Residuals

- Prediction: plug in X to get predicted Y :

$$\hat{Y} = b_0 + b_1 X$$

- Residual = Actual Y – Predicted $\hat{Y}$
- Residual plots help assess model fit
 - Look for patterns $\Rightarrow$ model may be inappropriate
 - No pattern $\Rightarrow$ model may be a good fit





Predictions and Residuals for Specific Temperatures

- Regression equation:

$$\hat{Y} = -0.29 + 0.23 \times \text{Temperature}$$

- At 70°F:

$$\hat{Y} = -0.29 + 0.23 \times 70 = 15.81 \quad \Rightarrow \quad \text{Predicted chirps}$$

- At 90°F:

$$\hat{Y} = -0.29 + 0.23 \times 90 = 20.41 \quad \Rightarrow \quad \text{Predicted chirps}$$

- Residuals = Actual chirps – Predicted chirps for these temperatures:

- At 70°F: Actual chirps (from data) ≈ 14.7 , residual = $14.7 - 15.81 = -1.11$
- At 90°F: Actual chirps (from data) ≈ 19.8 , residual = $19.8 - 20.41 = -0.61$

Summary: Key Ideas

- Simple linear regression models a straight-line relationship
- Slope tells us how Y changes per unit increase in X
- Intercept is the predicted value of Y when $X = 0$
- Line chosen to minimize squared errors: least squares
- Predictions and residuals help evaluate fit